

Figure 1 — Three-Tier Epistemic Architecture with Canonical Bridge Principles
(Paper 4, §3.2)

Three-Tier Epistemic Architecture

Tier 3 — Computational

Currency: prediction errors ϵ , precision Π ,
variational F, d', AUC
Evidence: single-trial EEG, model fit, BIC/AIC

Tier 2 — Information-Theoretic

Currency: bits, nats, mutual information,
channel capacity, Φ
Evidence: transfer entropy, bias-corrected MI, Φ

Tier 1 — Thermodynamic

Currency: joules, ATP, watts, entropy
Evidence: ^{31}P -MRS, two-photon imaging,
BOLD-calibrated fMRI (proxy)

Valid bridges (solid)
Absent bridges (dashed $\times$)

T2→T3
Not yet specified $\times$

T3→T1: Double bridge (3→2→1)
requires both above
Status: composite, unvalidated

T1→T2
Not yet specified $\times$

T3→T2: κ bridge (ATP/bit)
approx. inference → MI optimisation
Status: grounded, κ unmeasured

T2→T1: Landauer bridge
 $E \geq kT \cdot \ln 2$ per bit erased
Status: theoretically bounded, κ unmeasured

Double-bridge
calculation

